

Brac University
Department of Electrical & Electronic Engineering
Fall2025

Course Number : EEE282

Section :01

Group No : 04



Lab Report

Experiment no: 05

Name of the experiment:

Curve Fitting: Linear & Polynomial Regression

Prepared by:

Name: Alif Tamjid ID: 24121308

All Group members:

<i>Sl.</i>	<i>ID</i>	<i>Name</i>
1.	24121205	Souvik Barman Ratul
2.	24121308	Alif Tamjid
3.	23221082	Sabrina Maliha
4.	24121019	Monazzi Siddique

Objective:

The objective of this experiment is to perform curve fitting using linear and polynomial regression to analyze discrete data. It aims to estimate intermediate values, derive simplified functions and minimize errors using least square regression.

Theoretical Background:

Curve fitting is a technique used to find a mathematical function that best represents a set of data. In linear regression, the goal is to fit a straight line to the data, minimizing the sum of squared errors between observed and predicted values. Polynomial regression on the other hand, fits a curve by using higher degree polynomials. It is used when the data show a non linear relationship. Both methods rely on the least squares approach to minimize errors and provide accurate predictions for intermediate values.

Linear Regression: Linear regression models the relationship between a dependent variable y and independent variable x . The formula is:

$$y = mx + c;$$

Here,

m = slope

c = intercept

The residual error is the difference between the observed and predicted values.

$$e = Y - y = Y - mx - c.$$

To minimize error of e , we can have a sum of square errors here,

$$S = \sum_{i=1}^N e_i^2 = \sum_{i=1}^N (Y_i - y_i)^2 = \sum_{i=1}^N (mx_i + c - y_i)^2$$

$$= \sum_{i=1}^N (m^2 x_i^2 + c^2 + y_i^2 + 2mcx_i - 2mx_i y_i - 2cy_i)$$

$$= m^2 \sum x_i^2 + Nc^2 + \sum y_i^2 + 2mc \sum x_i - 2m \sum x_i y_i - 2c \sum y_i \quad \text{--- (1)}$$

To determine the values of m and c , eq (1) differentiated with respect to each co-efficient and it is equal to zero.

$$\frac{\partial S}{\partial m} = 2m \sum x_i^2 + 2c \sum x_i - 2 \sum x_i y_i = 0$$

$$\frac{\partial S}{\partial c} = 2Nc + 2m \sum x_i - 2 \sum y_i = 0$$

$$\therefore m = \frac{N \sum x_i y_i - \sum x_i \sum y_i}{N \sum x_i^2 - (\sum x_i)^2}, \quad c = \frac{1}{N} (\sum y_i - m \sum x_i)$$

$$= \bar{y} - m\bar{x}$$

Figure 1

Polynomial Regression: Polynomial Regression is used to model the relationship between dependent variable y and an independent variable x using the formula,

$$y = a_2 * (x)^2 + a_1 * x + a_0;$$

Here, a_2 = quadric coefficient, a_1 = linear coefficient and a_0 = intercept value

For this, the sum of the square of residue is, $S = \sum_{i=1}^N e_i^2 = \sum_{i=1}^N (y_i - \hat{y}_i)^2$
 $= \sum_{i=1}^N (y_i - a_0 - a_1 x_i - a_2 x_i^2)^2 \quad \text{--- (ii)}$

Taking derivation of eq (ii) with respect to unknown coefficient a_0 , a_1 and a_2 and it is equal to zero $\rightarrow$

$$\frac{\partial S}{\partial a_0} = -2 \sum (y_i - a_0 - a_1 x_i - a_2 x_i^2) = 0$$

$$\frac{\partial S}{\partial a_1} = -2 \sum x_i (y_i - a_0 - a_1 x_i - a_2 x_i^2)$$

$$\frac{\partial S}{\partial a_2} = -2 \sum x_i^2 (y_i - a_0 - a_1 x_i - a_2 x_i^2) = 0$$

By solving these,

$$Na_0 + (\sum x_i) a_1 + (\sum x_i^2) a_2 = \sum y_i$$

$$(\sum x_i) a_0 + (\sum x_i^2) a_1 + (\sum x_i^3) a_2 = \sum x_i y_i$$

$$(\sum x_i^2) a_0 + (\sum x_i^3) a_1 + (\sum x_i^4) a_2 = \sum x_i^2 y_i$$

a_0 , a_1 and a_2 can be calculated using matrix inversion.

Figure 2

Problem Statement:

Problem-1: The following data shows the relationship between the viscosity of SAE 70 oil and temperature.

Temperature (°C)	26.67	93.33	148.89	315.56
Viscosity (N.s/m ²)	1.35	0.085	0.012	0.00075

(a) Plot the data-points in MATLAB and decide whether a straight line could fit them.

(b) After taking the log of the data, use linear regression to find the equation of the line that best fits the data.

Answer:**Code:**

(a)

```
clear all;
close all;
clc;
x = [26.67 93.33 148.69 315.56]; %x = Temperature in
Degree Celcius
y = [1.35 0.085 0.012 0.00075]; %y = Viscosity in
(N.s/m2)
```

```
%Question(a): Plot the data-points in MATLAB and
decide whether a straight line could fit them.
```

```
%Plotting the points
```

```
figure(1)
plot(x,y, 'ro', 'linewidth', 2);
```

```
%Displaying the name of the axes
```

```
xlabel('Temperature');
ylabel('Viscosity');
grid on;
hold on;
```

Output:

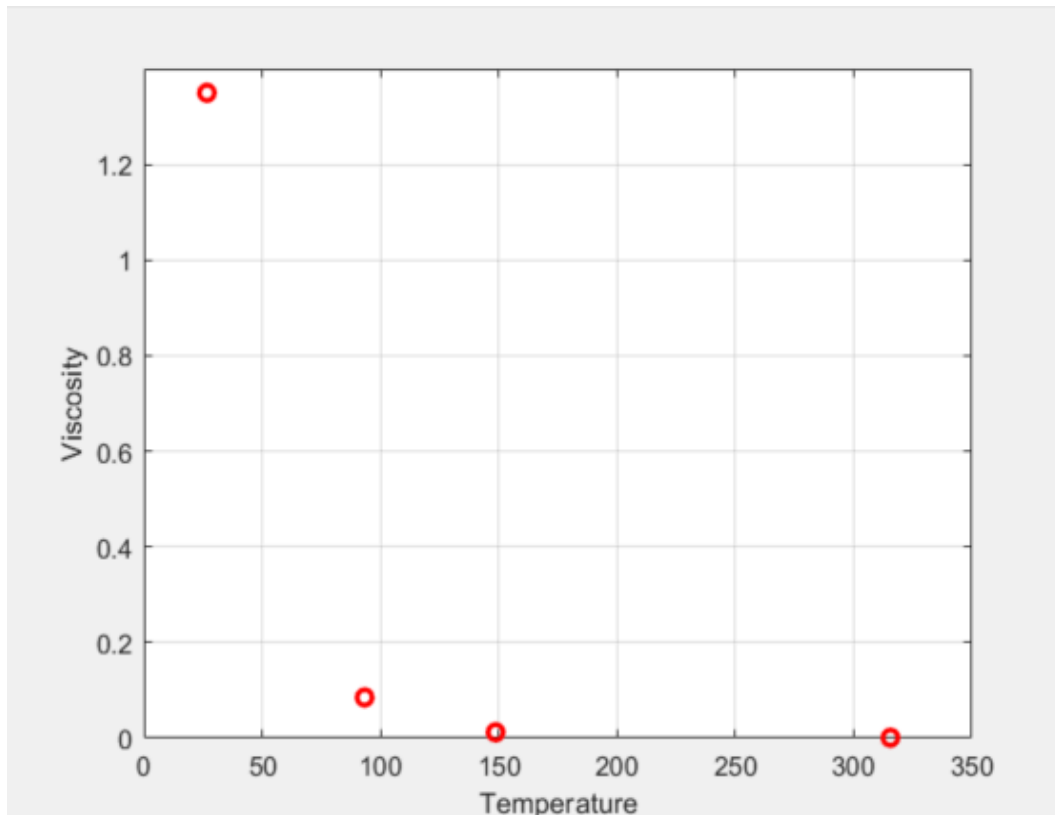


Figure 3: Points plotted in the Viscosity vs temperature graph.

Explanation:

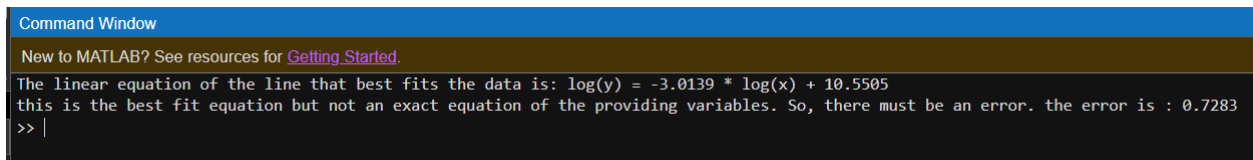
In this code we plot these data points on a graph using MATLAB to check if a straight line could fit the data. The x-axis represents temperature (°C) and the y-axis represents viscosity (N·s/m²). A grid and labels are added for better visualization of the relationship between temperature and viscosity. From the plotted points, it is clear that a straight line cannot fit this data accurately.

(b)

Code:

```
clear all;
close all;
clc;
x = [26.67 93.33 148.69 315.56]; %x = Temperature in
Degree Celcius
y = [1.35 0.085 0.012 0.00075]; %y = Viscosity in
(N.s/m^2)
%Question(b): After taking the log of the data, use
linear regression to find the equation of the line
that best fits the data.
%calculating the logarithm of the given value
x1 = log(x);
y1 = log(y);
% Find out the value of N, m and c which is needed to
find a liner equation. Linear equation:  $y = mx + c$ 
N = length(x1);
m = (N.*sum(x1.*y1)-sum(x1).*sum(y1))/(N.*sum(x1.^2)-(sum(x1)).^2);
c = mean(y1)- m.*mean(x1);
fprintf('The linear equation of the line that best
fits the data is:  $\log(y) = 0.4f * \log(x) + 0.4f$ \n', m,
c);
%find out the error
Y_func = @(xf) m.*xf + c;
Y = Y_func(x1);
S = sum((Y-y1).^2);
fprintf('This is the best fit equation but not an
exact equation of the providing variables. So, there
must be an error. the error is : 0.4f\n', S);
```

Output:



```
Command Window
New to MATLAB? See resources for Getting Started.
The linear equation of the line that best fits the data is: log(y) = -3.0139 * log(x) + 10.5505
this is the best fit equation but not an exact equation of the providing variables. So, there must be an error. the error is : 0.7283
>> |
```

Figure 4: Log(Viscosity) vs Log(Temperature) linear regression best-fit line.

Explanation:

In this code, the logarithm of temperature and viscosity values is taken to linearize the data. Linear regression is then used to find the slope and intercept of the best-fit line. The resulting equation shows the relationship between log(viscosity) and log(temperature). Although the fitted line closely represents the data trend, it is not an exact equation, so a small error value is calculated to show the difference between the actual and predicted data.

Discussion:

From the plotted data, it is observed that viscosity decreases sharply as temperature increases, showing an inverse relationship between the two variables. In part (a), the data points did not follow a straight-line pattern, indicating that a simple linear relationship could not represent the data accurately. Therefore, in part (b), the logarithm of both temperature and viscosity was taken to linearize the relationship. The linear regression on the log-transformed data provided a best-fit equation with a small error, confirming that the log-log relationship better represents the experimental trend. Overall, the analysis shows that viscosity of SAE 70 oil decreases exponentially with increasing temperature.

Problem-2: The data below represents the bacterial growth in a liquid culture over a number of days

Day	0	4	8	12	16	20
Amount (x 10 ⁶)	67	84	98	125	149	185

Find a best-fit equation to the data trend. Try several possibilities:

(a) linear (1st order)

(a) parabolic (2nd order)

(b) predict the number of bacteria after 40 days for both of the fitted equations.

Answer:

Code:

(a+b) For liner(1st order):

```
clear all;  
close all;  
clc;
```

```
% Problem:02(a)- Find a best-fit equation to the data  
trend for linear (1st order)
```

```
% Input the variables
```

```
x = 0:4:20; % x = Days  
y = [67 84 98 125 149 185]; % y = Amount(x106)
```

```
% Find out the value of N, m and c which is needed to  
find a linear equation. Linear equation:  $y = mx + c$ 
```

```
N = length(x);
```

```

m      =      (N.*      sum(x.*y) -      sum(x) .*sum(y)) /
(N.*sum(x.^2) - (sum(x)).^2);
c = mean(y) - m.*mean(x);

```

%putting a function so that we can measure every point precisely for finding a linear equation

```

Y_func = @(xf) m.*xf + c;

```

```

xr = 0 : 0.01 : 20;
yr = Y_func(xr);

```

% best fit linear equation figure

```

figure(1);
plot(xr, yr, 'b');
grid on;
hold on;

```

```

fprintf('The best fit linear equation is: y = %.4f * x
+ %.4f\n', m, c);

```

% In this figure, we can see a figure which is a straight line. From this prove, we can reach to a conclusion that we are successful to find the best fit linear equation which is $y = mx + c$

% I know this is a best fit equation but not an exact equation of the providing variables. So, there must be an error. Let's find the error

```

Y = Y_func(x);
S = sum((Y-y).^2);

```

```

fprintf('Error = %.4f\n', S);

```

```
% (b) predict the number of bacteria after 40 days for  
the fitted equations
```

```
day = 40;  
bacteria_numbers = Y_func(day);
```

```
fprintf('the number of bacteria after 40 days for the  
fitted equation is: %d', bacteria_numbers);
```

Output:

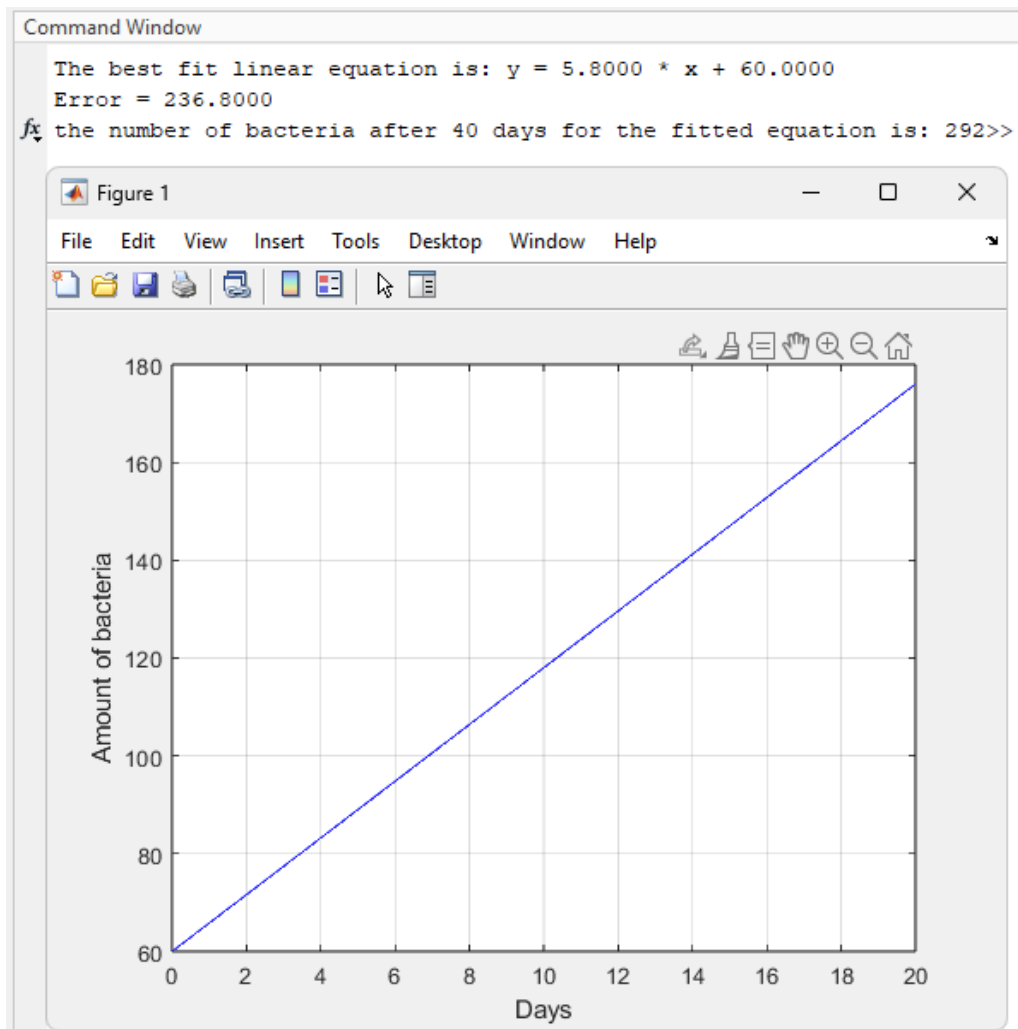


Figure 5: Best-fit Linear Regression line for bacterial growth data.

Explanation:

In this code, the value of days and amount of the bacterial growth has been taken to linearize the data. Firstly, Linear regression is being used to find out the slope and intercept value of the best fit linear line. After that, the resulting equation comes which shows the relationship between the days (x) and the amount of bacteria (y). It represents the best fit linear equation of this data which is visualized in the above output's figure. We also predict the number of bacteria after 40 days from the best fitted linear equation.

Discussion:

From the plotted graph, we can see that the amount of bacteria is increasing with respect to the days which are also increasing. It represents a proportional relation to the bacterial growth in a liquid culture over a number of days. Moreover, it indicates a best fit linear relationship between them. We also identified the value of bacteria after 40 days from the linear equation from the best fitted equation.

(a+b) For parabolic(2nd order):

```
clear all;
close all;
clc;

% Problem:02(a)- Find a best-fit equation to the data
trend for parabolic (2nd order)

% Input the variables

x = 0:4:20; % x = Days
y = [67 84 98 125 149 185]; % y = Amount(x106)

% the parabolic equation is:  $a(x^2) + bx + c = 0$ . To
find the best fit of this equation, let's distribute
it in a  $PA = Q$  where we have to find out the matrix of
A.

P = [length(x) sum(x) sum(x.^2);
      sum(x) sum(x.^2) sum(x.^3);
      sum(x.^2) sum(x.^3) sum(x.^4)];
Q = [sum(y) sum(x.*y) sum((x.^2).*y)]';

% Let's find the matrix A

A = P\Q; % A = [Ao A1 A2]

fprintf('The best fit parabolic equation is: %.4f *
(x^2) + %.4f * x + %.4f = 0\n', A(3), A(2), A(1));

%putting a function so that we can measure every point
precisely for finding a parabola equation

Y_func= @(xf) A(3).*(xf).^2 + A(2).*xf + A(1);
```

```

xr = 0 : 0.01 : 20;
yr = Y_func(xr);

% best fit parabola equation figure

figure(1);
plot(xr, yr, 'r');
xlabel('Days');
ylabel('Amount of bacteria');
grid on;
hold on;

% In this figure, we can see a figure which is a
parabola. From this prove, we can reach out to a
conclusion that we are successful to find the best fit
parabola equation which is  $a(x^2) + bx + c = 0$ 

% I know this is a best fit equation but not an exact
equation of the providing variables. So, there must be
an error. Let's find the error

Y = Y_func(x);
S = sum((Y-y).^2);

fprintf('Error = %.4f\n', S);

% (b) predict the number of bacteria after 40 days for
the fitted equations

day = 40;
bacteria_numbers = Y_func(day);

fprintf('the number of bacteria after 40 days for the
fitted equation is: %d', bacteria_numbers);

```

Output:

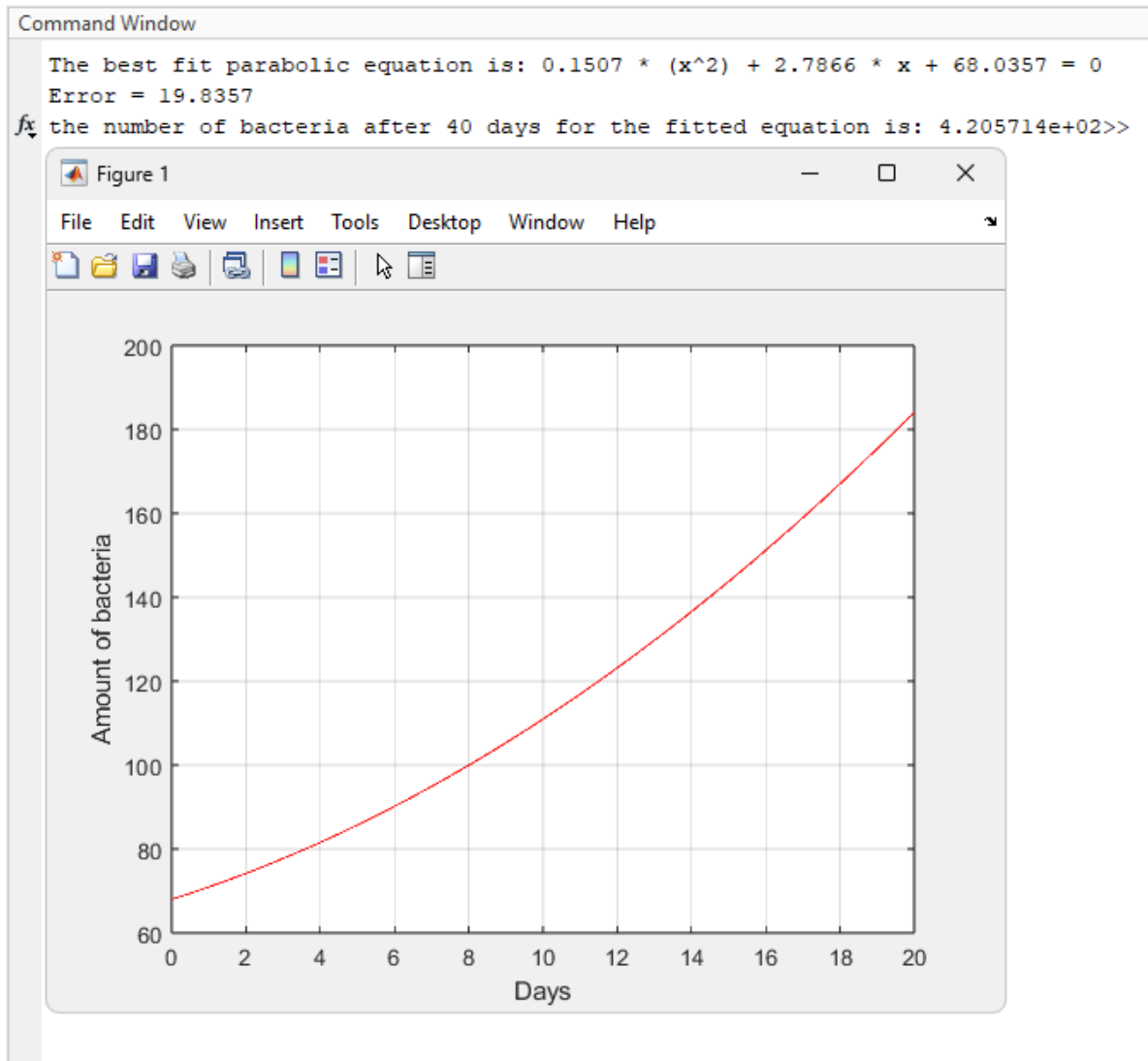


Figure 6: Best-fit Parabolic (2nd order) Regression curve for bacterial growth data

Explanation:

In this code, the value of days and amount of the bacterial growth has been taken to parabolize the data. Firstly, parabolic regression is being used to find out the quadric coefficient which indicates growth acceleration, linear coefficient which indicates initial growth rate and intercept value which indicates starting bacterial amount of the best fit parabolic line. After that, the resulting equation comes which shows the relationship between the days (x) and the amount of bacteria (y). It represents the best fit parabolic equation of this data which is visualized in the above output's figure. We also predict the number of bacteria after 40 days from the best fitted linear equation.

Discussion:

From the plotted graph, we can see that the bacterial population increases faster with each passing day. It represents the growth rate is not constant but accelerating, which is typical for bacterial cultures during the exponential phase. The fitted equation helps in predicting population size and understanding growth dynamics. We also identified the value of bacteria after 40 days from the linear equation from the best fitted equation.

Problem-3: Consider the following data table:

x	0.4	0.8	1.2	1.6	2	2.3
y	800	975	1500	1950	2900	3600

To fit these data points, which type of equation do you think would be suitable?
Find the equation accordingly

Answer:

Code:

```
clear all;
```

```
close all;
```

```
clc;
```

```
x = [0.4 0.8 1.2 1.6 2 2.3];
```

```
y = [800 975 1500 1950 2900 3600];
```

```
% Question(c)- To fit these data points, which type of  
equation do you think would be suitable? Find the  
equation accordingly.
```

```
%Plotting the points
```

```
figure(1)
```

```

plot(x,y, 'ro', 'linewidth', 2);

%Displaying the name of the axes

xlabel('x');
ylabel('y');

grid on;
hold on;

fprintf('For linear equation:\n\n');

% Find out the value of N, m and c which is needed to
find a liner equation. Linear equation:  $y = mx + c$ 

N = length(x);

m = (N.*sum(x.*y) - sum(x).*sum(y)) /
(N.*sum(x.^2) - (sum(x)).^2);

c = mean(y) - m.*mean(x);

fprintf('The best fit linear equation is:  $y = %.4f * x$ 
+  $%.4f$ \n', m, c);

%find out the error of linear equation

Y_func = @(xf) m.*xf + c;

```

```

Y = Y_func(x);
S1 = sum((Y-y).^2);

fprintf('The error for the linear equation is:
%.4f\n\n\n', S1);

fprintf('For parabolic equation:\n\n');

% the parabolic equation is:  $a(x^2) + bx + c = 0$ . To
find the best fit of this equation, let's distribute
it in a  $PA = Q$  where we have to find out the matrix of
A.

P = [length(x) sum(x) sum(x.^2);
      sum(x) sum(x.^2) sum(x.^3);
      sum(x.^2) sum(x.^3) sum(x.^4)];
Q = [sum(y) sum(x.*y) sum((x.^2).*y)]';

% Let's find the matrix A

A = P\Q; % A = [Ao A1 A2]

fprintf('The best fit parabolic equation is: %.4f *
(x^2) + %.4f * x + %.4f = 0\n', A(3), A(2), A(1));

```

```
%Find out the error for the parabolic equation
```

```
Z_func= @(xf) A(3).*(xf).^2 + A(2).*xf + A(1);
```

```
Z = Z_func(x);
```

```
Sp = sum((Z-y).^2);
```

```
fprintf('Error = %.4f\n\n\n', Sp);
```

```
if Sp>S1
```

```
    fprintf('As Sp>S1, the error of linear equation is  
less than parabolic equation. So, The best equation is  
linear equation.\n\n');
```

```
    fprintf('The best fit equation is: y = %.4f * x +  
%.4f\n', m, c);
```

```
    xr = 0.4:0.01:2.3;
```

```
    yr = Y_func(xr);
```

```
    figure(2)
```

```
    plot(xr, yr, 'r');
```

```
    grid on;
```

```
    hold on;
```

```
else
```

```
fprintf('As  $S_1 > S_p$ , the error of parabolic equation  
is less than linear equation. So, The best equation is  
parabolic equation.\n\n');
```

```
fprintf('The best fit equation is: %.4f * (x^2) +  
%.4f * x + %.4f = 0\n', A(3), A(2), A(1));
```

```
xr = 0.4:0.01:2.3;
```

```
yr = Z_func(xr);
```

```
figure(2)
```

```
plot(xr, yr, 'r');
```

```
xlabel('x');
```

```
ylabel('y');
```

```
grid on;
```

```
hold on;
```

```
end
```

Output:

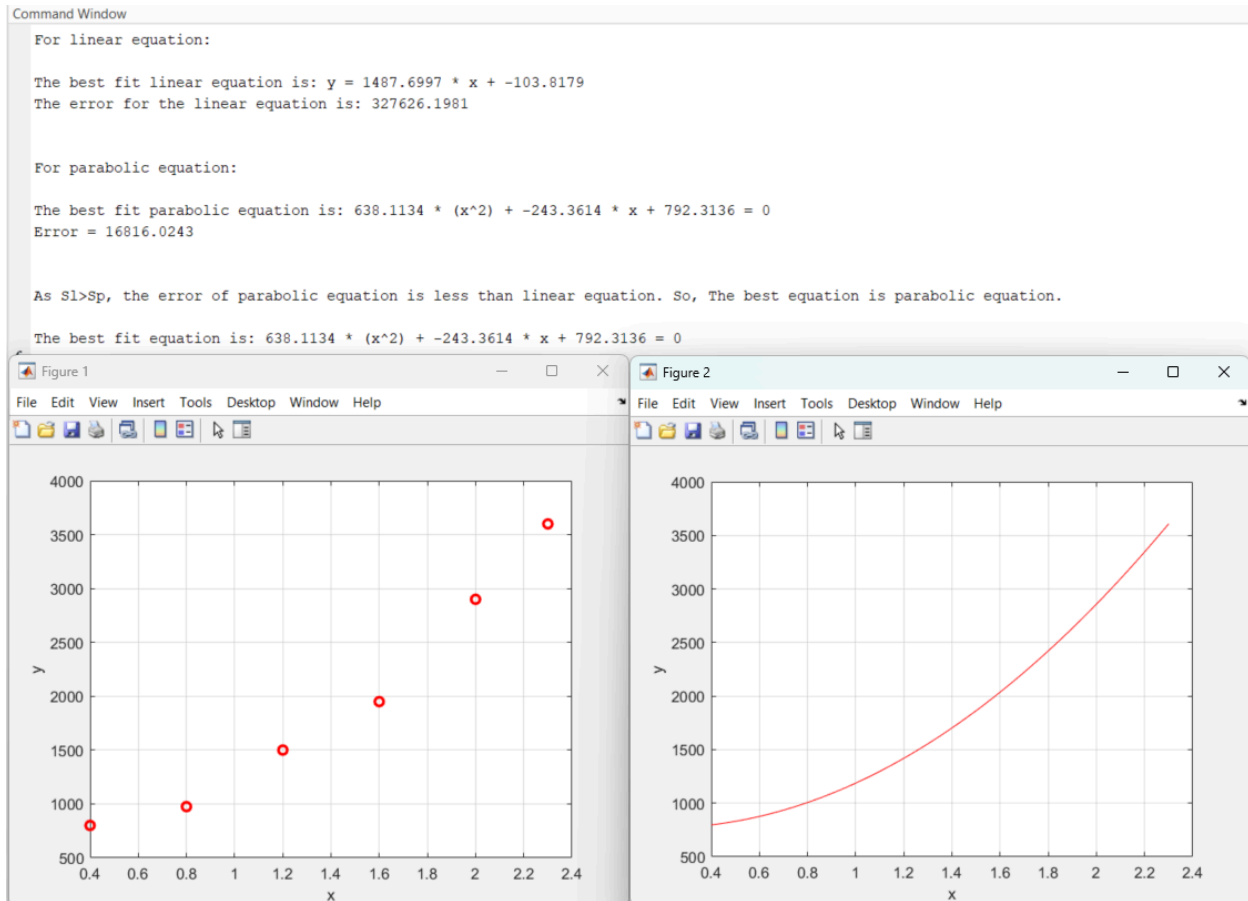


Figure 7: Parabolic best-fit curve for the relationship between x and y.

Explanation:

In this code, the given data values of x and y are analyzed to determine which equation best fits the data trend. Both linear and parabolic regressions were applied, and their respective errors were compared. The parabolic equation showed a smaller error than the linear one, indicating a better fit. Therefore, the relationship between x and y is represented by a parabolic equation. The plotted figure visually confirms that the data points align closely with the parabolic fit.

Discussion:

From the graphical output, it is observed that the data follows an almost parabolic pattern, suggesting it is not constant but accelerating in y with respect to x . This indicates a parabolic relationship where the change in y is faster than x . The smaller error value further supports the accuracy of the parabolic model over the linear one. Hence, the parabolic regression equation provides the most reliable prediction and explanation for the given data trend.